Logic-Gated Biomaterials for Targeted Therapy

Exploring how "Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery" is changing delivery methods for bioactive therapeutics

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The history of hydrogels and their applications to drug delivery

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What current challenges do hydrogels face in clinical applications?

03 **Innovation**

How did one group of researchers innovate biomaterial technology to solve these challenges?

Background

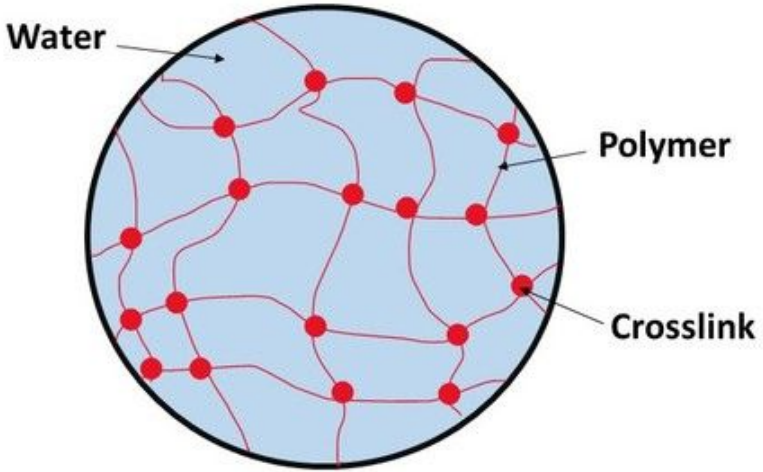
Biomaterial of Interest: Hydrogels

What are hydrogels?

- A 3D network of insoluble, hydrophilic polymers
- Has the ability to hold large amounts of water while maintaining its structure

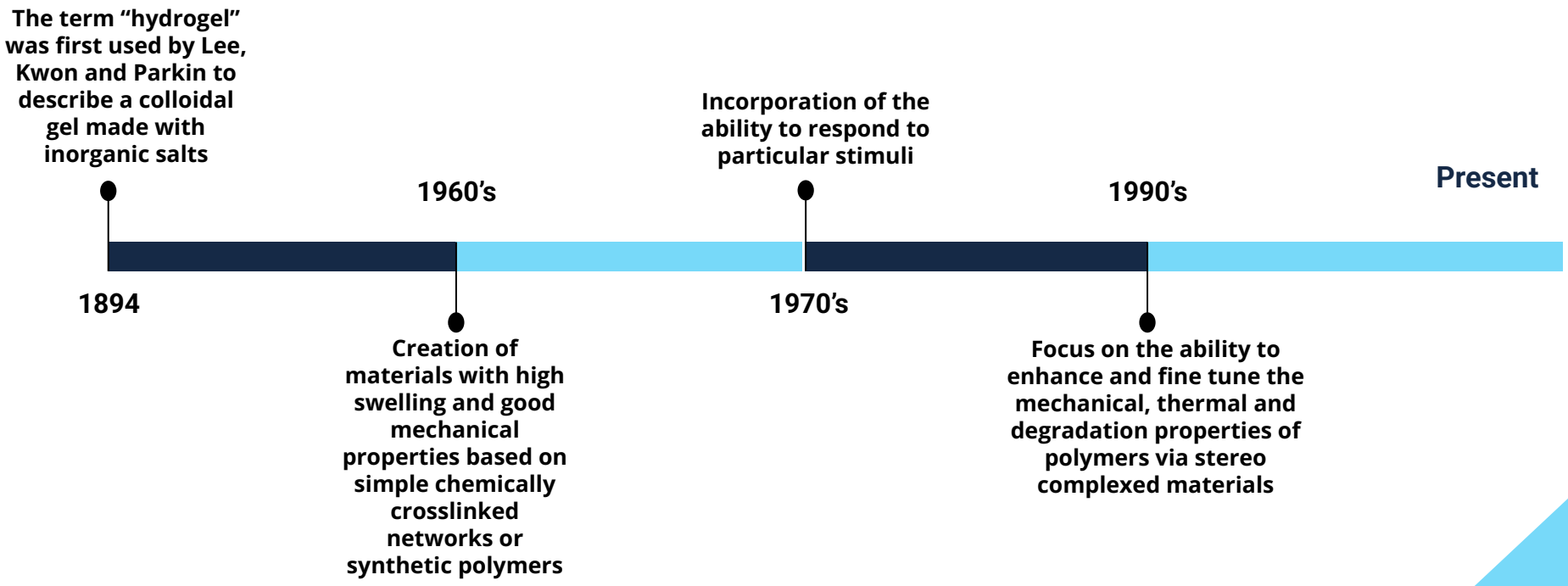
Classes

- Natural and synthetic



Slutz, S. (2023, October 12). Can Biodegradable Hydrogels Help Conserve Water in Farming? Retrieved from https://www.sciencebuddies.org/science-fair-projects/project_ideas/AgTech_p013/agricultural-technology/water-conservation-hydrogels

Generations of Hydrogels



Present: Smart Hydrogels as Drug Delivery Systems

- The properties of the previous generations of hydrogels have been tailored to provide promising results in stability and release kinetics, making smart hydrogels a potential solution for targeted drug delivery.

Bordbar-Khiabani, A., & Gasik, M. (2022, March 27). *Smart hydrogels for Advanced Drug Delivery Systems.* MDPI. https://www.mdpi.com/1422-0067/23/7/3665



SPAAC-Based PEG Hydrogels

(strain-promoted azide-alkyne cycloaddition)

Cancer: Hydrogel technology could stop tumors from returning. (2022, April 19).

https://www.medicalnewstoday.com/articles/hydrogel-patches-could-prevent-melanoma-breast-cancer-and-ovarian-cancer-tumors-from-returning

Properties of SPAAC-Based PEG Hydrogels



Cross-linking

SPAAC cross-linking is selective and efficient between functional groups



Rapid Gelation

The fast kinetics of the formation process leads to an efficient gellation



Versatility

The chemical structure of PEG can be easily modified



Tunability

The mechanical properties of PEG hydrogels can be easily tuned



Biocompatibility

The hydrophilicity and non-toxic degradation of hydrogels promote biocompatibility

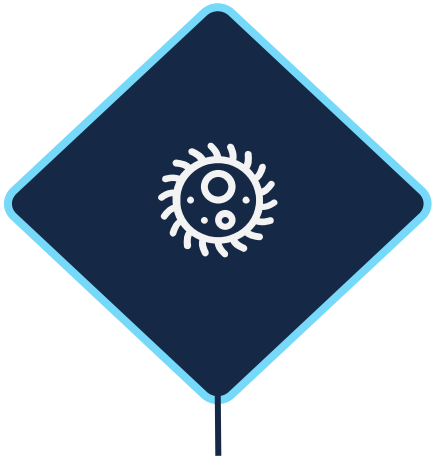
Clinical Challenges

Limitations

Release Specificity

Uniform response mechanisms lack the ability to trigger release.





Microenvironments

Complex diseases have dynamic and heterogeneous biological conditions that hydrogels cannot adapt to

Combination Therapy

Conventional hydrogels do not allow for sequential or time dependent combination therapy



Clinical Potential of More Advanced Hydrogels

01

Diagnostics

Enable precise disease marker detection



02

Drug Delivery

Enhance targeted and controlled release profiles



03

Cell Culture

Provide 3D environment for tailored cell responses



04

Regenerative Tech

Responsive cell guidance for precise tissue regeneration



Next-step Proposal

Challenge of Targeted Therapeutic Delivery

Precision of delivering therapeutics to areas affected by disease, and designing delivery mechanisms that are responsive to more than one biological input

Technology Overview



01

Biocomputation Capability

- Sense multiple inputs
- Execute Boolean logic-based algorithms

02

Modular Design

- Allows for alteration and combination of inputs and logic functions

03

Cytocompatibility

- Formed and degraded in the presence of live cells without causing harm
- Stimuli sensitive components

Results

Cross Linkers

Successfully synthesized **17** distinct cross-linkers each demonstrating a unique logic output when exposed to different combinations of enzyme, reductant, and light.
- tested with **MALDI–TOF**

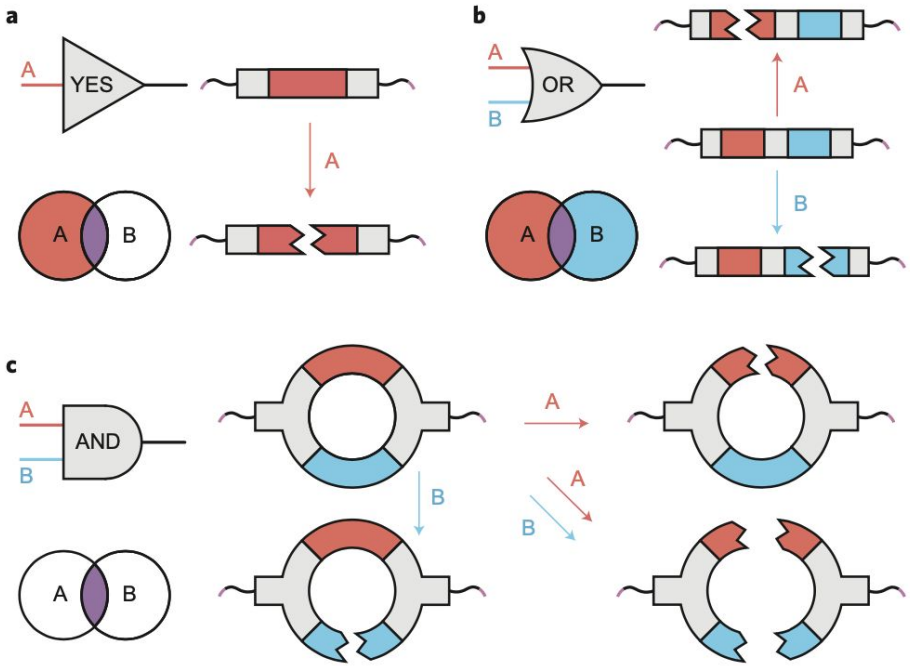


Figure 1a, b and c. Badeau, B. A., Comerford, M. P., Arakawa, C. K., Shadish, J. A., & DeForest, C. A. (2018). Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery. *Nature Chemistry*, *10*(3), Article 3. https://doi.org/10.1038/nchem.2917

Disease- associated delivery

Hydrogel designed to degrade and release DOX in response to specific cancer microenvironment cues

Absence of treatment or with single input:
dsDNA content of **95 ± 3%, 97 ± 2%, and 76 ± 5%**

With both relevant inputs (MMP-8 and reductant):
dsDNA content of **1.8 ± 0.2%**

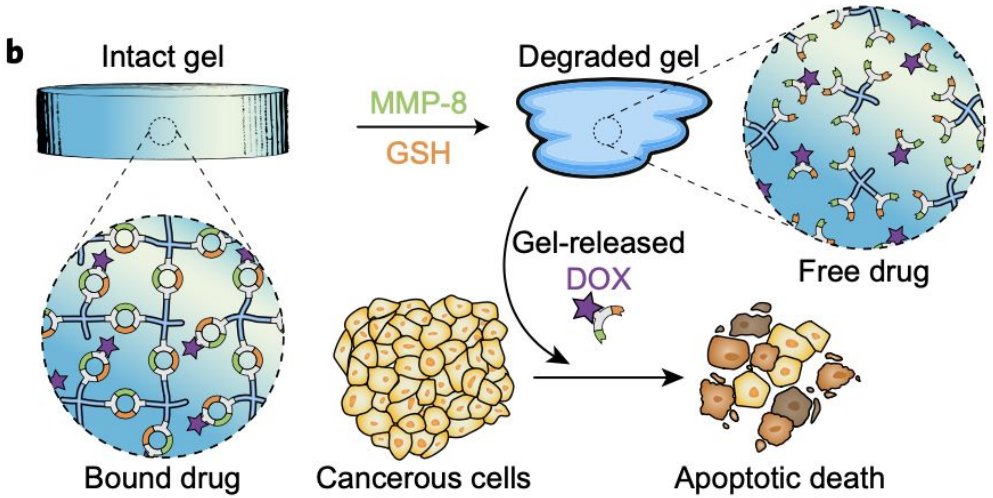


Figure 4b. Badeau, B. A., Comerford, M. P., Arakawa, C. K., Shadish, J. A., & DeForest, C. A. (2018). Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery. *Nature Chemistry*, *10*(3), Article 3. https://doi.org/10.1038/nchem.2917

Sequential and Spatiotemporally Varied Delivery

Hydrogel designed with segregated regions, each labeled with a different fluorophore

Each of these regions engineered to respond to a particular set of environmental conditions or stimuli, such as light or chemical reducers

When exposed to these specific stimuli, only the targeted region within the hydrogel degrade

Supplementary Figure 11. Badeau, B. A., Comerford, M. P., Arakawa, C. K., Shadish, J. A., & DeForest, C. A. (2018). Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery. *Nature Chemistry*, *10*(3), Article 3. https://doi.org/10.1038/nchem.2917

Mechanical Properties

Precisely controlled by manipulating environmental stimuli

Pretreating the E∧P linker with different combinations of enzyme and light and adding it to PEG to test formation of hydrogel

Untreated:
G′ = 1,660 ± 170 Pa

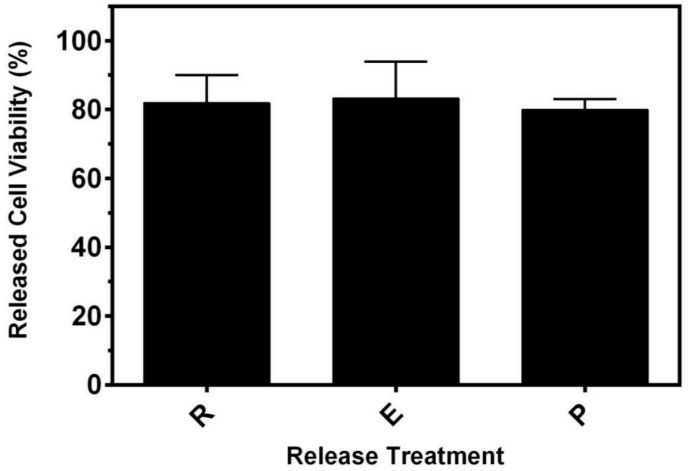
Treated with Enzyme or Light:
G′ = 1,580 ± 130 Pa and 1,540 ± 110 Pa

Treated with both Enzyme and Light:
G′ = 200 ± 30

Cytocompatibility

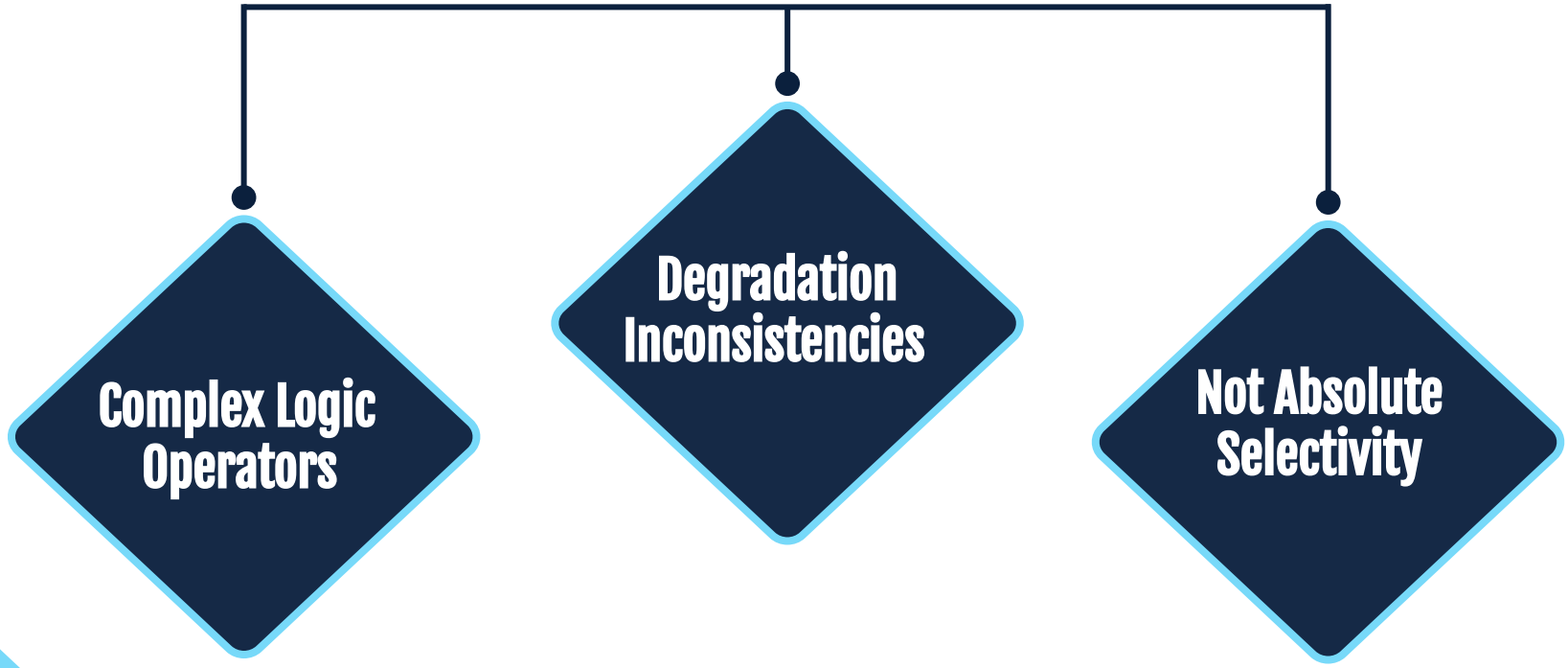
Encapsulated cells remained viable after being released

Significant advancement in live-cell delivery platforms



Supplementary Figure 12 Badeau, B. A., Comerford, M. P., Arakawa, C. K., Shadish, J. A., & DeForest, C. A. (2018). Engineered modular biomaterial logic gates for environmentally triggered therapeutic delivery. *Nature Chemistry*, *10*(3), Article 3. https://doi.org/10.1038/nchem.2917

Limitations of Technology



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Questions?

Thank you to Dr. Bansal and the TAs for a wonderful semester!